

ddr_version_detect Module

Design Report

Training-Based DDR Generation Fingerprinting — DDR2 / DDR3 / DDR4 / DDR5

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§1 Problem Statement

A DFI 5.2 PHY exposes only electrical signals — `DQ[63:0]`, `DQS`, `CA[13:0]`, `CK`, `CS#`, `CKE`, `RESET#` — with no sideband that identifies which DDR generation is physically seated. The same 288-pin DIMM connector is shared from DDR2 through DDR5. The Memory Controller (MC) therefore cannot statically configure its timing parameters or command encoder at synthesis time.

Two detection strategies exist:

1. **SPD (Serial Presence Detect)** — I²C EEPROM on the DIMM carries an authoritative version byte. Fast (< 100 cycles), but requires an I²C master in the MC.
2. **Training-based fingerprint** — MC issues generation-specific training commands, measures responses, and matches against per-version templates. Works even when SPD is absent or unreadable. This is the method implemented in `ddr_version_detect`.

SPD can be added as a fast-path bypass that skips the probe FSM when a valid SPD byte is available.

§2 Key Insight: Per-Generation Timing Identity

Every DDR generation has a unique combination of timing parameters and protocol features that can be observed without writing data to the array. These form a fingerprint vector that the module reads during the training window.

Parameter	DDR2	DDR3	DDR4	DDR5
Min CL (cycles)	3	5	9	32
Write Levelling	—	Yes	Yes	Yes
CA bus type	SE	SE	SE	Diff
MRS register count	4	4	7	8+
VREF training	—	—	DQ+CA	On-die
DFE training	—	—	—	Yes
t_{WL} min (ns)	—	7.5	7.5	4.0

Table 1: Per-generation distinguishing parameters. No two rows are identical.

The CA bus type (differential vs. single-ended) is the strongest single discriminator: a DDR5 differential CA probe fails hard on DDR4/3/2 with no ambiguity.

§3 Module Interface

Detection runs once at init; output is held until the next reset. Runtime re-detection is possible by re-asserting `start_detect`.

Port	Dir	Description
<code>clk</code>	in	System clock
<code>rst_n</code>	in	Active-low reset
<code>start_detect</code>	in	Assert to begin detection; re-assertable at runtime
<code>phy_dq[63:0]</code>	in	Raw DQ samples from PHY
<code>phy_dqs</code>	in	DQS strobe
<code>phy_ca[13:0]</code>	in	CA bus samples
<code>phy_ca_diff</code>	in	High if PHY detects differential CA activity (DDR5)
<code>training_ack</code>	in	PHY ACK pulse after training command
<code>ddr_version[2:0]</code>	out	Detected generation: 0=DDR2, 1=DDR3, 2=DDR4, 3=DDR5
<code>detected</code>	out	Strobe when <code>ddr_version</code> is valid
<code>confidence[3:0]</code>	out	Match score from comparator (0=low, 15=high)
<code>train_params[N]</code>	out	Packed CL, t_{WL} , MRS count, etc. for MC config mux

Table 2: Module port list.

§4 Block Diagram

Figure 1 shows the internal structure of `ddr_version_detect`. Data flows left-to-right: PHY signals enter, training probes are issued by the FSM, measurements feed the comparator, and the best-match result is latched into the Output Register.

§5 Internal Block Descriptions

5.1 Reset & Init Seq

Drives the DRAM through the JEDEC power-on reset sequence common to all DDR generations: `CKE` held low, `RESET#` pulse, and a stabilisation window. Asserts `rst_done` to the Probe FSM when it is safe to begin probing.

5.2 Probe FSM

The control core. Iterates probes in order `DDR5` $\rightarrow$ `DDR4` $\rightarrow$ `DDR3` $\rightarrow$ `DDR2`, issuing generation-specific `probe_cmd` signals to both the Timing Meas Unit and the MRS Response Checker simultaneously. After all probes complete, it triggers the Score Comparator and forwards the lock signal to the Output Register.

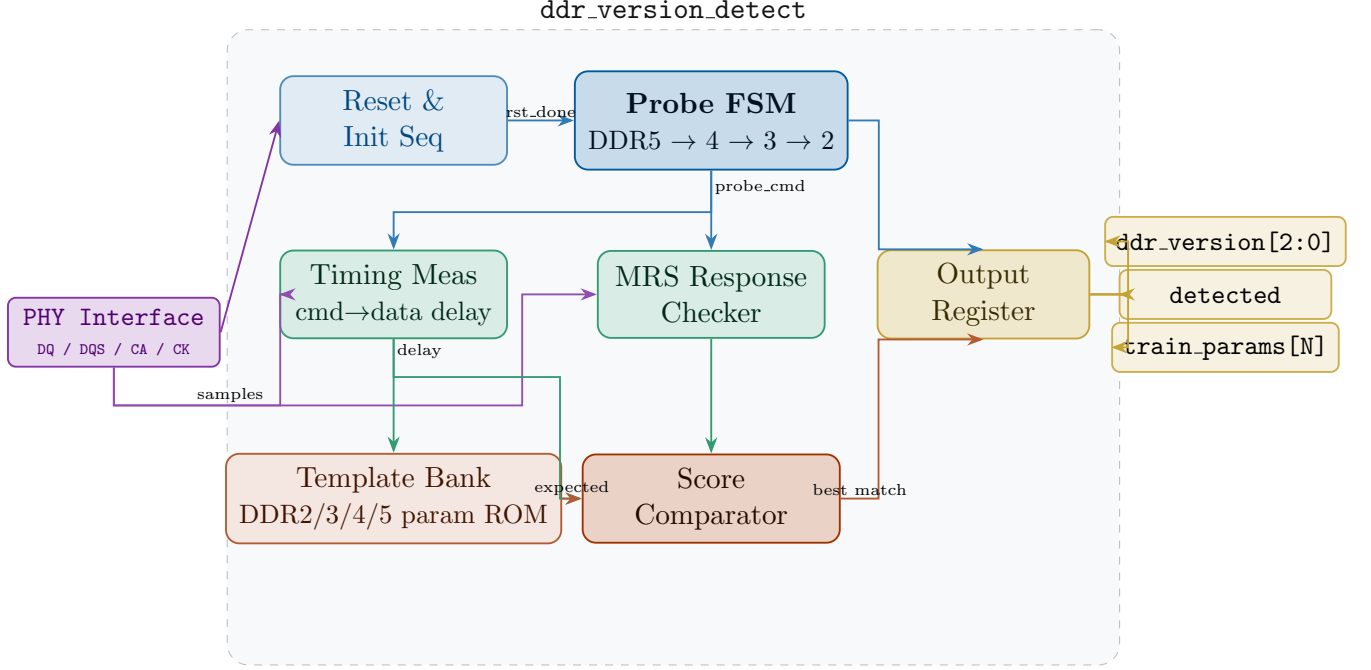


Figure 1: Top-level block diagram of `ddr_version_detect`. **Purple**: PHY input; **Blue**: FSM/control; **Green**: measurement; **Orange**: template matching; **Yellow**: output.

5.3 Timing Measurement Unit

Counts clock cycles from command issue (`probe_cmd`) to when valid data appears on DQ. The `cmd→data` delay window is a hard fingerprint:

Generation	CL range (cycles)
DDR2	3–6
DDR3	5–11
DDR4	9–24
DDR5	32+

Table 3: CL ranges have no overlap between generations.

5.4 MRS Response Checker

Verifies whether the DRAM acknowledges a Mode Register Write correctly. DDR5 defines 8+ MR addresses; DDR4 defines 7; DDR3/2 define 4. Sending a DDR5 MR encoding to a DDR3 part yields no ACK or a garbage response, producing a mismatch score.

5.5 Template Bank

A read-only parameter store holding one expected feature vector per generation:

$$\mathbf{v}_{\text{gen}} = [CL_{\min}, CL_{\max}, t_{WL}, \text{MRS_count}, \text{CA_type}, \text{VREF_support}]$$

Indexed by generation ID (0–3), output on `expected` to the Score Comparator each probe cycle.

5.6 Score Comparator

Computes a weighted distance between the measured vector $\hat{\mathbf{v}}$ and each template $\mathbf{v}_g$:

$$\text{dist}_g = w_{\text{CL}} \cdot |\Delta \text{CL}| + w_{\text{CA}} \cdot \mathbf{1}[\text{CA_mismatch}] + w_{t_{\text{WL}}} \cdot |\Delta t_{\text{WL}}| + \dots$$

The CA bus type carries the highest weight w_{CA} because a DDR5 differential CA probe fails categorically on all other generations. The generation with minimum dist_g is selected as **best match**.

5.7 Output Register

Latches the winning generation and associated parameters on the **lock** strobe from the Probe FSM. Drives **ddr_version**[2:0], **detected**, **confidence**[3:0], and the packed **train_params** bus to the MC config mux downstream.

§6 Probe Sequences

Each probe sends a generation-specific training command and observes the response:

Probe	Command sent	Observable measured
DDR5	MRW on differential CA bus, JEDEC MR0 encoding	Diff CA response + $t_{\text{WL}} = 4 \text{ ns}$ window + on-die VREF ACK
DDR4	MRS0 (CL=12) single-ended CA; VREFDQ training cmd	CL=12 valid eye + VREF sweep converges + $t_{\text{WL}} \approx 7.5 \text{ ns}$
DDR3	MRS0 (CL=7); write-levelling enable via MRS1 bit A7	DQS skew response; CL 5–11 valid; no VREF cmd ACK
DDR2	EMRS1 OCD calibration command	ZQ response without MRS3/4+; CL 3–6; no write-levelling

Table 4: Per-generation probe sequences and measured observables.

Fallback Rule

The DDR5 differential CA probe fails hard on DDR4/3/2 due to bus topology mismatch. This single binary outcome is the strongest discriminator and is weighted highest in the Score Comparator.

§7 Probe FSM State Diagram

